

LECTURE 0: INTRODUCTION – COURSE OVERVIEW

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Whom am I

- Mingyi Hong, Assistant Professor at Iowa State University, Industrial Engineering Department
- Main research area

First-order optimization theory and methods

Applications in Signal Processing

Applications in Networking and Communications

Applications in Machine Learning

About this short course

- The title is: “Non-Convex Decomposition Algorithms and Applications in Big Data”
- The objective is to introduce an number of recent first-order algorithms for solving a few emerging problems
- We will mostly focus on **non-convex** algorithms
- Our focus will be on **decomposition** algorithms
- We will touch on applications in machine learning, signal processing and communications

Main algorithms to be covered

- We will cover a few algorithms, given below

The Block Successive Upper Bound Minimization (BSUM) algorithm

The Block Coordinate Descent (BCD) algorithm

The Alternating Direction Method of Multipliers (ADMM) [both convex and non-convex versions]

The Stochastic Gradient Descent (SGD) based algorithms

Maybe more...

Main references (tutorials)

- S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, “Distributed optimization and statistical learning via the alternating direction method of multipliers,” *Foundations and Trends in Machine Learning*, vol. 3, no. 1, pp. 1 - 122
- M. Hong, M. Razaviyayn, Z.-Q. Luo, and J.-S. Pang, “A unified algorithmic framework for block structured optimization involving big data,” *IEEE Signal Processing Magazine*, vol. 33, no. 1, pp. 57 - 77, 2016.
- H. Baligh, M. Hong, W.-C. Liao, Z.-Q. Luo, M. Razaviyayn, M. Sanjabi, and R. Sun, “Cross-layer provision of future cellular networks: A WMMSE-based approach,” *IEEE Signal Processing Magazine*, vol. 31, no. 6, pp. 56 - 68, Nov 2014

Main references (algorithms)

- R. Johnson and T. Zhang, “Accelerating stochastic gradient descent using predictive variance reduction,” in the Proceedings of the Neural Information Processing (NIPS). 2013
- M. Razaviyayn, M. Hong, and Z.-Q. Luo, “A unified convergence analysis of block successive minimization methods for nonsmooth optimization,” SIAM Journal on Optimization, vol. 23, no. 2, pp. 1126 - 1153, 2013.
- P. Richtarik and M. Takac, “Iteration complexity of randomized block-coordinate descent methods for minimizing a composite function,” Mathematical Programming, vol. 144, pp. 1-38, 2014.
- M. Hong, X. Wang, M. Razaviyayn, and Z.-Q. Luo, “Iteration complexity analysis of block coordinate descent methods,” 2013

Schedule (temporary)

